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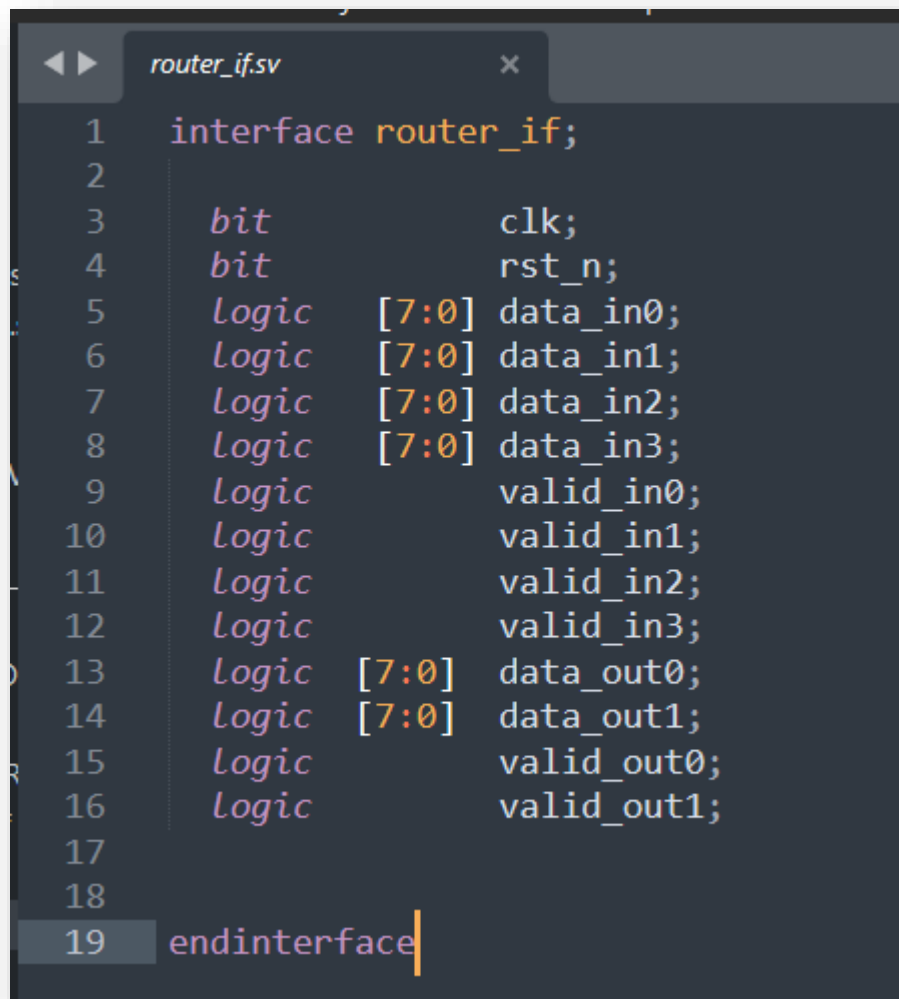
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🔗 *The environment will just have*

- Function new for uvm objects and uvm components
 - Build phase for uvm components ONLY
-

1. UVM verification environment:

1. Router Interface



```
router_if.sv
1  interface router_if;
2
3      bit          clk;
4      bit          rst_n;
5      logic [7:0]  data_in0;
6      logic [7:0]  data_in1;
7      logic [7:0]  data_in2;
8      logic [7:0]  data_in3;
9      logic        valid_in0;
10     logic        valid_in1;
11     logic        valid_in2;
12     logic        valid_in3;
13     logic [7:0]  data_out0;
14     logic [7:0]  data_out1;
15     logic        valid_out0;
16     logic        valid_out1;
17
18
19  endinterface
```

2. Router Package

```
1 ▼ package router_pkg;|
2
3     `include "router_transaction.sv"
4     `include "router_sequence.sv"
5     `include "router_test.sv"
6     `include "router_enviroment.sv"
7
8     `include "router_scoreboard.sv"
9     `include "router_coverage.sv"
10    `include "router_agent.sv"
11
12    `include "router_sequencer.sv"
13    `include "router_driver.sv"
14    `include "router_monitor.sv"
15
16 endpackage
```

3. Top module

```
1  import uvm_pkg::*;
2  include "uvm_macros.svh"
3
4      import router_pkg::*;
5      `include "router_if.sv"
6
7  module top;
8
9      router_if routerif();
10
11     // DUT Instantiation ::
12     router dut (
13         .clk (routerif.clk),
14         .rst_n (routerif.rst_n),
15         .data_in0 (routerif.data_in0),
16         .data_in1 (routerif.data_in1),
17         .data_in2 (routerif.data_in2),
18         .data_in3 (routerif.data_in3),
19         .valid_in0 (routerif.valid_in0),
20         .valid_in1 (routerif.valid_in1),
21         .valid_in2 (routerif.valid_in2),
22         .valid_in3 (routerif.valid_in3),
23         .data_out0 (routerif.data_out0),
24         .data_out1 (routerif.data_out1),
25         .valid_out0 (routerif.valid_out0),
26         .valid_out1 (routerif.valid_out1)
27     );
28
29     // Clock generation ::
30     initial begin
31         routerif.clk = 0;
32         forever begin
33             #2 routerif.clk = ~routerif.clk;
34         end
35     end
```

```
35     end
36
37     // Rest ::
38     initial begin
39         routerif.rst_n = 0;
40         #2 routerif.rst_n = 1;
41     end
42
43     // RUN TEST ::
44     initial begin
45         //uvm_top.set_report_verbosity_level(UVM_HIGH);
46         $display("DEFAULT VERRBOSITY LEVEL IS %0d ",uvm_top.get_report_verbosity_level);
47         run_test("router_test");
48     end
49
50 endmodule
```

4. Router Transaction (Sequence items)

Note : The macros define is used in all the modules / files

```
1 | #ifndef ROUTER_TRANSACTION_SVH
2 | #define ROUTER_TRANSACTION_SVH // to avoid multiple inclusion issues , if its included it will not define it
3 |
4 | import uvm_pkg::*;
5 | `include "uvm_macros.svh"
6 |
7 | class router_transaction extends uvm_sequence_item;
8 |
9 |     // Define Signals
10 |
11 |     // Inputs to randomize ::
12 |     rand logic [7:0] data_in0;
13 |     rand logic [7:0] data_in1;
14 |     rand logic [7:0] data_in2;
15 |     rand logic [7:0] data_in3;
16 |     rand logic      valid_in0;
17 |     rand logic      valid_in1;
18 |     rand logic      valid_in2;
19 |     rand logic      valid_in3;
20 |
21 |     // Outputs ::
22 |     logic [7:0] data_out0;
23 |     logic [7:0] data_out1;
24 |     logic      valid_out0;
25 |     logic      valid_out1;
26 |
27 | /*-----
28 | Factory Registration ::
29 | - registerate class in factory so it can be created, to make UVM see it so when I need to make operations on it , I could
30 | - Tells UVM how to construct objects of class via factory , enabling reusability and override
31 | -----*/
32 |
33 | `uvm_object_utils_begin(router_transaction)
34 |
35 |     // to be able to use uvm_object methods in processing my data ::
36 |     `uvm_field_int( data_in0, UVM_DEFAULT)
37 |     `uvm_field_int( data_in1, UVM_DEFAULT)
38 |     `uvm_field_int( data_in2, UVM_DEFAULT)
39 |     `uvm_field_int( data_in3, UVM_DEFAULT)
40 |     `uvm_field_int( valid_in0, UVM_DEFAULT)
41 |     `uvm_field_int( valid_in1, UVM_DEFAULT)
42 |     `uvm_field_int( valid_in2, UVM_DEFAULT)
43 |     `uvm_field_int( valid_in3, UVM_DEFAULT)
44 |
45 | `uvm_object_utils_end
46 |
47 | // Function new , which Actually creates the Object ::
48 | function new(string name = "router_transaction");
49 |     super.new(name);
50 |     $display("\n ===== SEQUENCE ITEM NEW =====");
51 |     `uvm_info("SEQUENCE_item","creating new sequence_item class (L)",UVM_LOW)
52 |     `uvm_info("SEQUENCE_item","creating new sequence_item class (M)",UVM_MEDIUM)
53 |     `uvm_info("SEQUENCE_item","creating new sequence_item class (H)",UVM_HIGH)
54 |     `uvm_info("SEQUENCE_item","creating new sequence_item class (FULL)",UVM_FULL)
55 |     $display("===== \n");
56 |
57 | endfunction
58 |
59 | endclass
60 |
61 | `endif
```

5. Router Sequence

```
1  `ifndef ROUTER_SEQUENCE_SVH
2  `define ROUTER_SEQUENCE_SVH
3
4      import uvm_pkg::*;
5      `include "uvm_macros.svh"
6
7      class router_sequence extends uvm_sequence;
8          `uvm_object_utils(router_sequence)
9
10         function new(string name = "router_sequence");
11             super.new(name);
12
13             $display("\n ===== SEQUENCE NEW =====");
14             `uvm_info("SEQUENCE","creating new sequence class (L)",UVM_LOW)
15             `uvm_info("SEQUENCE","creating new sequence class (M)",UVM_MEDIUM)
16             `uvm_info("SEQUENCE","creating new sequence class (H)",UVM_HIGH)
17             `uvm_info("SEQUENCE","creating new sequence class (FULL)",UVM_FULL)
18             $display("===== \n");
19         endfunction
20
21     endclass
22
23 `endif
```

6. Router Test

```
1  `ifndef ROUTER_TEST_SVH
2  `define ROUTER_TEST_SVH
3
4      import uvm_pkg::*;
5      `include "uvm_macros.svh"
6
7      `include "router_enviroment.sv"
8      `include "router_sequence.sv"
9
10     class router_test extends uvm_test;
11         `uvm_component_utils(router_test)
12
13         // For the building phase ::
14         router_env env;
15         router_sequence seq;
16
17         // For creating new test classes ::
18         function new(string name = "router_test", uvm_component parent);
19             super.new(name, parent);
20             $display("\n ===== TEST NEW =====");
21             `uvm_info("TEST","creating new TEST class (L)",UVM_LOW)
22             `uvm_info("TEST","creating new TEST class (M)",UVM_MEDIUM)
23             `uvm_info("TEST","creating new TEST class (H)",UVM_HIGH)
24             `uvm_info("TEST","creating new TEST class (FULL)",UVM_FULL)
25             $display("===== \n");
26
27         endfunction
```

```

28
29     function void build_phase (uvm_phase phase);
30         super.build_phase(phase);
31         env = router_env::type_id::create("env", this);
32         seq = router_sequence::type_id::create("seq",this);
33
34         $display("\n ===== TEST BUILD =====");
35         `uvm_info(get_type_name()," Inside router TEST build phase (L)",UVM_LOW)
36         `uvm_info(get_type_name()," Inside router TEST build phase (M)",UVM_MEDIUM)
37         `uvm_info(get_type_name()," Inside router TEST build phase (H)",UVM_HIGH)
38         `uvm_info(get_type_name()," Inside router TEST build phase (FULL)",UVM_FULL)
39         $display("===== \n");
40
41     endfunction
42 endclass
43
44 `endif

```

7. Router Environment

```

1  |ifndef ROUTER_ENV_SVH
2  |define ROUTER_ENV_SVH
3
4      import uvm_pkg::*;
5      `include "uvm_macros.svh"
6
7      `include "router_agent.sv"
8      `include "router_coverage.sv"
9      `include "router_scoreboard.sv"
10
11
12      class router_env extends uvm_env;
13          `uvm_component_utils(router_env)
14
15          // For the building phase ::
16          router_agent agt;
17          router_scoreboard scb;
18          router_coverage cov;
19
20          // For creating new test classes ::
21          function new(string name = "router_env", uvm_component parent);
22              super.new(name, parent);
23
24              $display("\n ===== ENVIROMENT NEW =====");
25              `uvm_info("ENV","creating new ENV class (L)",UVM_LOW)
26              `uvm_info("ENV","creating new ENV class (M)",UVM_MEDIUM)
27              `uvm_info("ENV","creating new ENV class (H)",UVM_HIGH)
28              `uvm_info("ENV","creating new ENV class (FULL)",UVM_FULL)
29              $display("===== \n");
30          endfunction
31

```

```

32     function void build_phase (uvm_phase phase);
33         super.build_phase(phase);
34         agt = router_agent::type_id::create("agt", this);
35         scb = router_scoreboard::type_id::create("scb",this);
36         cov = router_coverage::type_id::create("cov",this);
37
38         $display("\n ===== ENVIROMENT BUILD =====");
39         `uvm_info(get_type_name()," Inside router ENV build phase (L)",UVM_LOW)
40         `uvm_info(get_type_name()," Inside router ENV build phase (M)",UVM_MEDIUM)
41         `uvm_info(get_type_name()," Inside router ENV build phase (H)",UVM_HIGH)
42         `uvm_info(get_type_name()," Inside router ENV build phase (FULL)",UVM_FULL)
43         $display("===== \n");
44
45     endfunction
46
47 endclass
48
49
50 `endif

```

8. Router Agent

```

1  `ifndef ROUTER_AGENT_SVH
2  `define ROUTER_AGENT_SVH
3
4      import uvm_pkg::*;
5      `include "uvm_macros.svh"
6
7      `include "router_sequencer.sv"
8      `include "router_driver.sv"
9      `include "router_monitor.sv"
10
11
12     class router_agent extends uvm_agent;
13         `uvm_component_utils(router_agent)
14
15         // For the building phase ::
16         router_sequencer sqr;
17         router_driver drv;
18         router_monitor mon;
19
20         // For creating new test classes ::
21         function new(string name = "router_agent", uvm_component parent);
22             super.new(name, parent);
23
24             $display("\n ===== AGENT NEW =====");
25             `uvm_info("agent","creating new agent class (L)",UVM_LOW)
26             `uvm_info("agent","creating new agent class (M)",UVM_MEDIUM)
27             `uvm_info("agent","creating new agent class (H)",UVM_HIGH)
28             `uvm_info("agent","creating new agent class (FULL)",UVM_FULL)
29             $display("===== \n");
30
31         endfunction

```

```

32
33     function void build_phase (uvm_phase phase);
34         super.build_phase(phase);
35         sqr = router_sequencer::type_id::create("sqr", this);
36         drv = router_driver::type_id::create("drv",this);
37         mon = router_monitor::type_id::create("mon",this);
38
39         $display("\n ===== AGENT BUILD =====");
40         `uvm_info(get_type_name()," Inside router AGENT build phase (L)",UVM_LOW)
41         `uvm_info(get_type_name()," Inside router AGENT build phase (M)",UVM_MEDIUM)
42         `uvm_info(get_type_name()," Inside router AGENT build phase (H)",UVM_HIGH)
43         `uvm_info(get_type_name()," Inside router AGENT build phase (FULL)",UVM_FULL)
44         $display("===== \n");
45     endfunction
46
47 endclass
48
49
50 `endif

```

9. Router Scoreboard

```

1  `ifndef ROUTER_SCOREBOARD_SVH
2  `define ROUTER_SCOREBOARD_SVH
3
4      import uvm_pkg::*;
5      `include "uvm_macros.svh"
6
7      class router_scoreboard extends uvm_scoreboard;
8          `uvm_component_utils(router_scoreboard)
9
10         function new(string name = "router_scoreboard", uvm_component parent);
11             super.new(name,parent);
12             $display("\n ===== SCOREBOARD NEW =====");
13             `uvm_info("SCOREBOARD","creating new SCOREBOARD class (L)",UVM_LOW)
14             `uvm_info("SCOREBOARD","creating new SCOREBOARD class (M)",UVM_MEDIUM)
15             `uvm_info("SCOREBOARD","creating new SCOREBOARD class (H)",UVM_HIGH)
16             `uvm_info("SCOREBOARD","creating new SCOREBOARD class (FULL)",UVM_FULL)
17             $display("===== \n");
18
19         endfunction
20
21         function void build_phase(uvm_phase phase);
22             super.build_phase(phase);
23             $display("\n ===== SCOREBOARD BUILD =====");
24             `uvm_info(get_type_name()," Inside router SCOREBOARD build phase (L)",UVM_LOW)
25             `uvm_info(get_type_name()," Inside router SCOREBOARD build phase (M)",UVM_MEDIUM)
26             `uvm_info(get_type_name()," Inside router SCOREBOARD build phase (H)",UVM_HIGH)
27             `uvm_info(get_type_name()," Inside router SCOREBOARD build phase (FULL)",UVM_FULL)
28             $display("===== \n");
29         endfunction
30
31     endclass
32
33
34 `endif

```

10. Router Coverage

```
1 `ifndef ROUTER_COVERAGE_SVH
2 `define ROUTER_COVERAGE_SVH
3
4 import uvm_pkg::*;
5 `include "uvm_macros.svh"
6
7 class router_coverage extends uvm_component;
8     `uvm_component_utils(router_coverage)
9
10     function new(string name = "router_coverage", uvm_component parent);
11         super.new(name,parent);
12
13         $display("\n ===== COVERAGE NEW =====");
14         `uvm_info("COVERAGE","creating new COVERAGE class (L)",UVM_LOW)
15         `uvm_info("COVERAGE","creating new COVERAGE class (M)",UVM_MEDIUM)
16         `uvm_info("COVERAGE","creating new COVERAGE class (H)",UVM_HIGH)
17         `uvm_info("COVERAGE","creating new COVERAGE class (FULL)",UVM_FULL)
18         $display("===== \n");
19
20     endfunction
21
22     function void build_phase(uvm_phase phase);
23         super.build_phase(phase);
24
25         $display("\n ===== COVERAGE NEW =====");
26         `uvm_info(get_type_name()," Inside router COVERAGE build phase (L)",UVM_LOW)
27         `uvm_info(get_type_name()," Inside router COVERAGE build phase (M)",UVM_MEDIUM)
28         `uvm_info(get_type_name()," Inside router COVERAGE build phase (H)",UVM_HIGH)
29         `uvm_info(get_type_name()," Inside router COVERAGE build phase (FULL)",UVM_FULL)
30         $display("===== \n");
31     endfunction
32
33 endclass
34
35 `endif
```

11. Router Sequencer

```
1 `ifndef ROUTER_SEQUENCER_SVH
2 `define ROUTER_SEQUENCER_SVH
3
4 import uvm_pkg::*;
5 `include "uvm_macros.svh"
6
7 class router_sequencer extends uvm_sequencer;
8     `uvm_component_utils(router_sequencer)
9
10     function new(string name = "router_sequencer", uvm_component parent);
11         super.new(name,parent);
12
13         $display("\n ===== SEQUENCER NEW =====");
14         `uvm_info("SQR","creating new SQR class (L)",UVM_LOW)
15         `uvm_info("SQR","creating new SQR class (M)",UVM_MEDIUM)
16         `uvm_info("SQR","creating new SQR class (H)",UVM_HIGH)
17         `uvm_info("SQR","creating new SQR class (FULL)",UVM_FULL)
18         $display("===== \n");
19     endfunction
20
21     function void build_phase(uvm_phase phase);
22         super.build_phase(phase);
23
24         $display("\n ===== SEQUENCER BUILD =====");
25         `uvm_info(get_type_name()," Inside router SQR build phase (L)",UVM_LOW)
26         `uvm_info(get_type_name()," Inside router SQR build phase (M)",UVM_MEDIUM)
27         `uvm_info(get_type_name()," Inside router SQR build phase (H)",UVM_HIGH)
28         `uvm_info(get_type_name()," Inside router SQR build phase (FULL)",UVM_FULL)
29         $display("===== \n");
30     endfunction
31 endclass
32
33 `endif
```

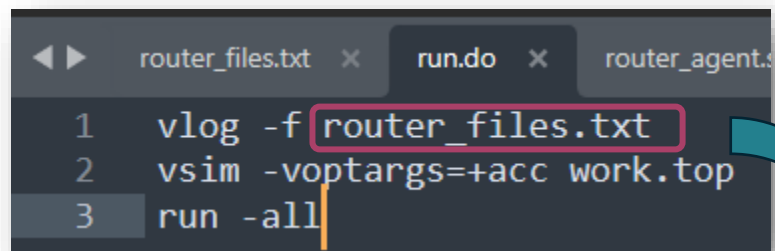
12. Router Driver

```
1 `ifndef ROUTER_DRIVER_SVH
2 `define ROUTER_DRIVER_SVH
3
4 `include "uvm_macros.svh"
5 import uvm_pkg::*;
6
7 class router_driver extends uvm_driver;
8   `uvm_component_utils(router_driver)
9
10   function new(string name = "router_driver", uvm_component parent);
11     super.new(name,parent);
12     $display("\n ===== DRIVER NEW =====");
13     `uvm_info("DRIVER","creating new DRIVER class (L)",UVM_LOW)
14     `uvm_info("DRIVER","creating new DRIVER class (M)",UVM_MEDIUM)
15     `uvm_info("DRIVER","creating new DRIVER class (H)",UVM_HIGH)
16     `uvm_info("DRIVER","creating new DRIVER class (FULL)",UVM_FULL)
17     $display("===== \n");
18
19   endfunction
20
21   function void build_phase(uvm_phase phase);
22     super.build_phase(phase);
23
24     $display("\n ===== DRIVER BUILD =====");
25     `uvm_info(get_type_name()," Inside router DRIVER build phase (L)",UVM_LOW)
26     `uvm_info(get_type_name()," Inside router DRIVER build phase (M)",UVM_MEDIUM)
27     `uvm_info(get_type_name()," Inside router DRIVER build phase (H)",UVM_HIGH)
28     `uvm_info(get_type_name()," Inside router DRIVER build phase (FULL)",UVM_FULL)
29     $display("===== \n");
30
31   endfunction
32 endclass
33
34 `endif
```

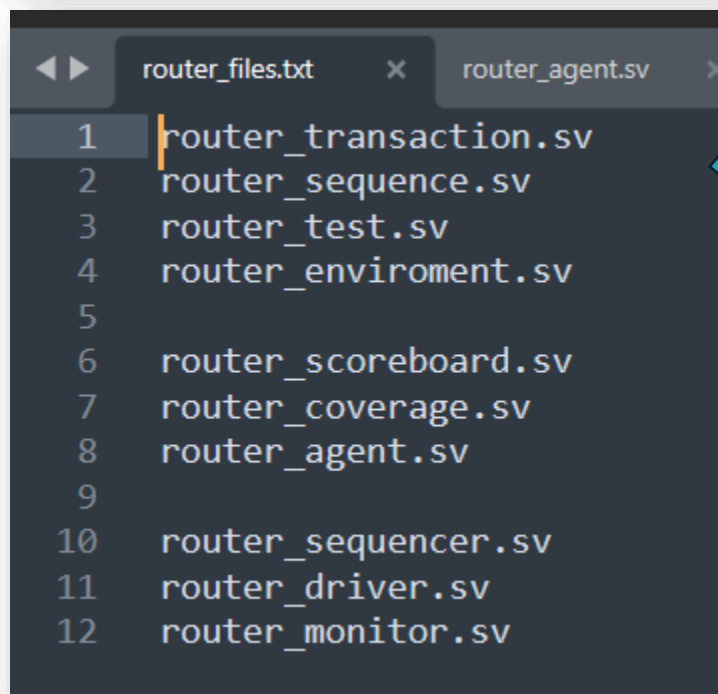
13. Router Monitor

```
1 `ifndef ROUTER_MONITOR_SVH
2 `define ROUTER_MONITOR_SVH
3
4 `include "uvm_macros.svh"
5 import uvm_pkg::*;
6
7 class router_monitor extends uvm_monitor;
8   `uvm_component_utils(router_monitor)
9
10   function new(string name = "router_monitor", uvm_component parent);
11     super.new(name,parent);
12     $display("\n ===== MONITOR NEW =====");
13     `uvm_info("MONITOR","creating new MONITOR class (L)",UVM_LOW)
14     `uvm_info("MONITOR","creating new MONITOR class (M)",UVM_MEDIUM)
15     `uvm_info("MONITOR","creating new MONITOR class (H)",UVM_HIGH)
16     `uvm_info("MONITOR","creating new MONITOR class (FULL)",UVM_FULL)
17     $display("===== \n");
18
19   endfunction
20
21   function void build_phase(uvm_phase phase);
22     super.build_phase(phase);
23
24     $display("\n ===== MONITOR NEW =====");
25     `uvm_info(get_type_name()," Inside router MONITOR build phase (L)",UVM_LOW)
26     `uvm_info(get_type_name()," Inside router MONITOR build phase (M)",UVM_MEDIUM)
27     `uvm_info(get_type_name()," Inside router MONITOR build phase (H)",UVM_HIGH)
28     `uvm_info(get_type_name()," Inside router MONITOR build phase (FULL)",UVM_FULL)
29     $display("===== \n");
30
31   endfunction
32 endclass
33
34
35 `endif
```

2. DO file for automating the process (Default verbosity)



```
1 vlog -f router_files.txt
2 vsim -voptargs=+acc work.top
3 run -all
```



```
1 router_transaction.sv
2 router_sequence.sv
3 router_test.sv
4 router_enviroment.sv
5
6 router_scoreboard.sv
7 router_coverage.sv
8 router_agent.sv
9
10 router_sequencer.sv
11 router_driver.sv
12 router_monitor.sv
```

3. Logs show all printed messages.

First thing, when running do file, it compiles all files and simulates

```
VSIM 12> do run.do
# QuestaSim-64 vlog 2021.1 Compiler 2021.01 Jan 19 2021
# Start time: 03:53:15 on Aug 13,2025
# vlog -reportprogress 300 -f router_files.txt
# ** Note: (vlog-2286) router_transaction.sv(5): Using implicit +incdir+D:/QuestaSim/uvim-1.1d/./verilog_src/uvim-1.1d/src from import uvim_pkg
# -- Compiling package router_transaction_sv_unit
# -- Importing package mtiUvm_uvm_pkg (uvim-1.1d Built-in)
# -- Compiling package router_sequence_sv_unit
# -- Compiling package router_test_sv_unit
# -- Compiling package router_enviroment_sv_unit
# -- Compiling package router_scoreboard_sv_unit
# -- Compiling package router_coverage_sv_unit
# -- Compiling package router_agent_sv_unit
# -- Compiling package router_sequencer_sv_unit
# -- Compiling package router_driver_sv_unit
# -- Compiling package router_monitor_sv_unit
#
# Top level modules:
# --none--
# End time: 03:53:16 on Aug 13,2025, Elapsed time: 0:00:01
# Errors: 0, Warnings: 0
# ** Error: 格数集和C数集構図
#
# Unable to replace existing ini file (D:/IC_2025/Hassan_Verification_2025_R1/Assignment/Assignmet_5_UVM_ROUTER/ASS_6_HK_R1_ROUTER_UVM.mpf). File can not be renamed.
# ** Error: 格数集和C数集構図
#
# Unable to replace existing ini file (D:/IC_2025/Hassan_Verification_2025_R1/Assignment/Assignmet_5_UVM_ROUTER/ASS_6_HK_R1_ROUTER_UVM.mpf). File can not be renamed.
# End time: 03:53:18 on Aug 13,2025, Elapsed time: 0:00:44
# Errors: 2, Warnings: 1
# vsim -voptargs="+acc" work.top
# Start time: 03:53:18 on Aug 13,2025
# ** Note: (vsim-8009) Loading existing optimized design _opt
# Loading sv_std.std
# Loading mtiUvm_uvm_pkg(fast)
# Loading work.router_pkg(fast)
# Loading work.top_sv_unit(fast)
# Loading mtiUvm.questa_uvm_pkg(fast)
# Loading work.top(fast)
# Loading work.router_if(fast_1)
# Loading work.router(fast)
# Loading D:/QuestaSim/uvim-1.1d\win64\uvim_dpi.dll
# -----
```

Then, it runs all and start printing (bytb3 ay3bt f l awl shwyt Info msh fahmaha :))

```
# -----
# UVM-1.1d
# (C) 2007-2013 Mentor Graphics Corporation
# (C) 2007-2013 Cadence Design Systems, Inc.
# (C) 2006-2013 Synopsys, Inc.
# (C) 2011-2013 Cypress Semiconductor Corp.
# -----
#
# ***** IMPORTANT RELEASE NOTES *****
#
# You are using a version of the UVM library that has been compiled
# with `UVM_NO_DEPRECATED undefined.
# See http://www.eda.org/svdb/view.php?id=3313 for more details.
#
# You are using a version of the UVM library that has been compiled
# with `UVM_OBJECT_MUST_HAVE_CONSTRUCTOR undefined.
# See http://www.eda.org/svdb/view.php?id=3770 for more details.
#
# (Specify +UVM_NO_RELNOTES to turn off this notice)
```

Then, it printed all New Functions UVM info and Build phase Functions:

Note: default verbosity is MEDIUM 😊

```
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(277) @ 0: reporter [Questa UVM] QUESTA_UVM-1.2.3
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(278) @ 0: reporter [Questa UVM]  questa_uvm::init(+struct)
# DEFAULT VERRBOSITY LEVEL IS 200
#
# ===== TEST NEW =====
# UVM_INFO router_test.sv(21) @ 0: uvm_test_top [TEST] creating new TEST class (L)
# UVM_INFO router_test.sv(22) @ 0: uvm_test_top [TEST] creating new TEST class (M)
# =====
#
# UVM_INFO @ 0: reporter [RNIST] Running test router_test...
#
# ===== ENVIROMENT NEW =====
# UVM_INFO router_enviroment.sv(25) @ 0: uvm_test_top.env [ENV] creating new ENV class (L)
# UVM_INFO router_enviroment.sv(26) @ 0: uvm_test_top.env [ENV] creating new ENV class (M)
# =====
#
# ===== SEQUENCE NEW =====
# UVM_INFO router_sequence.sv(14) @ 0: reporter@@router_sequence [SEQUENCE] creating new sequence class (L)
# UVM_INFO router_sequence.sv(15) @ 0: reporter@@router_sequence [SEQUENCE] creating new sequence class (M)
# =====
#
# ===== TEST BUILD =====
# UVM_INFO router_test.sv(35) @ 0: uvm_test_top [router_test]  Inside router TEST build phase (L)
# UVM_INFO router_test.sv(36) @ 0: uvm_test_top [router_test]  Inside router TEST build phase (M)
# =====
#
# ===== AGENT NEW =====
# UVM_INFO router_agent.sv(25) @ 0: uvm_test_top.env.agt [agent] creating new agent class (L)
# UVM_INFO router_agent.sv(26) @ 0: uvm_test_top.env.agt [agent] creating new agent class (M)
# =====
#
# ===== SCOREBOARD NEW =====
# UVM_INFO router_scoreboard.sv(13) @ 0: uvm_test_top.env.scb [SCOREBOARD] creating new SCOREBOARD class (L)
# UVM_INFO router_scoreboard.sv(14) @ 0: uvm_test_top.env.scb [SCOREBOARD] creating new SCOREBOARD class (M)
# =====
#
# ===== COVERAGE NEW =====
# UVM_INFO router_coverage.sv(14) @ 0: uvm_test_top.env.cov [COVERAGE] creating new COVERAGE class (L)
# UVM_INFO router_coverage.sv(15) @ 0: uvm_test_top.env.cov [COVERAGE] creating new COVERAGE class (M)
# =====
#
```

```

# ===== MONITOR NEW =====
# UVM_INFO router_monitor.sv(13) @ 0: uvm_test_top.env.agt.mon [MONITOR] creating new MONITOR class (L)
# UVM_INFO router_monitor.sv(14) @ 0: uvm_test_top.env.agt.mon [MONITOR] creating new MONITOR class (M)
#
#
# ===== AGENT BUILD =====
# UVM_INFO router_agent.sv(40) @ 0: uvm_test_top.env.agt [router_agent] Inside router AGENT build phase (L)
# UVM_INFO router_agent.sv(41) @ 0: uvm_test_top.env.agt [router_agent] Inside router AGENT build phase (M)
#
#
# ===== DRIVER BUILD =====
# UVM_INFO router_driver.sv(25) @ 0: uvm_test_top.env.agt.drv [router_driver] Inside router DRIVER build phase (L)
# UVM_INFO router_driver.sv(26) @ 0: uvm_test_top.env.agt.drv [router_driver] Inside router DRIVER build phase (M)
#
#
# ===== MONITOR NEW =====
# UVM_INFO router_monitor.sv(25) @ 0: uvm_test_top.env.agt.mon [router_monitor] Inside router MONITOR build phase (L)
# UVM_INFO router_monitor.sv(26) @ 0: uvm_test_top.env.agt.mon [router_monitor] Inside router MONITOR build phase (M)
#
#
# ===== SEQUENCER BUILD =====
# UVM_INFO router_sequencer.sv(25) @ 0: uvm_test_top.env.agt.sqr [router_sequencer] Inside router SQR build phase (L)
# UVM_INFO router_sequencer.sv(26) @ 0: uvm_test_top.env.agt.sqr [router_sequencer] Inside router SQR build phase (M)
#
#
# ===== COVERAGE NEW =====
# UVM_INFO router_coverage.sv(26) @ 0: uvm_test_top.env.cov [router_coverage] Inside router COVERAGE build phase (L)
# UVM_INFO router_coverage.sv(27) @ 0: uvm_test_top.env.cov [router_coverage] Inside router COVERAGE build phase (M)
#
#
# ===== SCOREBOARD BUILD =====
# UVM_INFO router_scoreboard.sv(24) @ 0: uvm_test_top.env.scb [router_scoreboard] Inside router SCOREBOARD build phase (L)
# UVM_INFO router_scoreboard.sv(25) @ 0: uvm_test_top.env.scb [router_scoreboard] Inside router SCOREBOARD build phase (M)
#
#

```

Finally, printed UVM Summary

```

# --- UVM Report Summary ---
#
# ** Report counts by severity
# UVM_INFO : 37
# UVM_WARNING : 0
# UVM_ERROR : 0
# UVM_FATAL : 0
# ** Report counts by id
# [COVERAGE] 2
# [DRIVER] 2
# [ENV] 2
# [MONITOR] 2
# [Questa UVM] 2
# [RNTST] 1
# [SCOREBOARD] 2
# [SEQUENCE] 2
# [SQR] 2
# [TEST] 2
# [agent] 2
# [router_agent] 2
# [router_coverage] 2
# [router_driver] 2
# [router_env] 2
# [router_monitor] 2
# [router_scoreboard] 2
# [router_sequencer] 2
# [router_test] 2
# ** Note: $finish : D:/QuestaSim/win64/./verilog_src/uvm-1.1d/src/base/uvm_root.svh(430)
# Time: 0 ns Iteration: 215 Instance: /top
# 1
# Break in Task uvm_pkg/uvm_root::run_test at D:/QuestaSim/win64/./verilog_src/uvm-1.1d/src/base/uvm_root.svh line 430
VSIM 3>

```

4. Reporting with different verbosity

We will use the method to do so:

1. Change the verbosity from do file switch
2. DO you remember the commented line in top module? will uncomment it 🧐

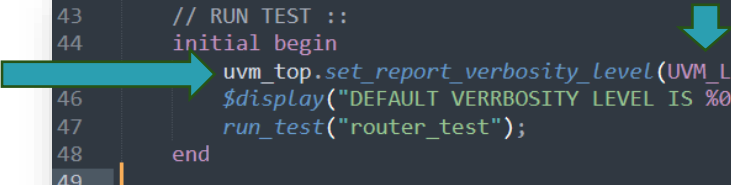
1. Low Verbosity

- DO file:

```
1 vlog -f router_files.txt
2 vsim -voptargs=+acc work.top +UVM_VERBOSITY=UVM_LOW
3 run -all
```

- Other option, in top module:

```
42
43 // RUN TEST ::
44 initial begin
45     uvm_top.set_report_verbosity_level(UVM_LOW);
46     $display("DEFAULT VERRBOSITY LEVEL IS %0d ",uvm_top.get_report_verbosity_level);
47     run_test("router_test");
48 end
49
50 endmodule
```



📌 Note:

If we change the verbosity level in the .do file and also call set_report_verbosity_level in the top module, the verbosity level set in the top module will take precedence.

(The top module setting will override the .do file)

- Logs:

```
# (Specify +UVM_NO_RELNOTES to turn off this notice)
#
# DEFAULT VERBOSITY LEVEL IS 100
#
# ===== TEST NEW =====
# UVM_INFO router_test.sv(21) @ 0: uvm_test_top [TEST] creating new TEST class (L)
# =====
#
# UVM_INFO @ 0: reporter [RNTST] Running test router_test...
#
# ===== ENVIROMENT NEW =====
# UVM_INFO router_enviroment.sv(25) @ 0: uvm_test_top.env [ENV] creating new ENV class (L)
# =====
#
# ===== SEQUENCE NEW =====
# UVM_INFO router_sequence.sv(14) @ 0: reporter@@router_sequence [SEQUENCE] creating new sequence class (L)
# =====
#
# ===== TEST BUILD =====
# UVM_INFO router_test.sv(35) @ 0: uvm_test_top [router_test] Inside router TEST build phase (L)
# =====
#
# ===== AGENT NEW =====
# UVM_INFO router_agent.sv(25) @ 0: uvm_test_top.env.agt [agent] creating new agent class (L)
# =====
#
# ===== SCOREBOARD NEW =====
# UVM_INFO router_scoreboard.sv(13) @ 0: uvm_test_top.env.scb [SCOREBOARD] creating new SCOREBOARD class (L)
# =====
#
# ===== COVERAGE NEW =====
# UVM_INFO router_coverage.sv(14) @ 0: uvm_test_top.env.cov [COVERAGE] creating new COVERAGE class (L)
# =====
#
# ===== ENVIROMENT BUILD =====
# UVM_INFO router_enviroment.sv(39) @ 0: uvm_test_top.env [router_env] Inside router ENV build phase (L)
# =====
#
# ===== SEQUENCER NEW =====
```

```

# ===== SEQUENCER NEW =====
# UVM_INFO router_sequencer.sv(14) @ 0: uvm_test_top.env.agt.sqr [SQR] creating new SQR class (L)
# =====
#
# ===== DRIVER NEW =====
# UVM_INFO router_driver.sv(13) @ 0: uvm_test_top.env.agt.drv [DRIVER] creating new DRIVER class (L)
# =====
#
# ===== MONITOR NEW =====
# UVM_INFO router_monitor.sv(13) @ 0: uvm_test_top.env.agt.mon [MONITOR] creating new MONITOR class (L)
# =====
#
# ===== AGENT BUILD =====
# UVM_INFO router_agent.sv(40) @ 0: uvm_test_top.env.agt [router_agent] Inside router AGENT build phase (L)
# =====
#
# ===== DRIVER BUILD =====
# UVM_INFO router_driver.sv(25) @ 0: uvm_test_top.env.agt.drv [router_driver] Inside router DRIVER build phase (L)
# =====
#
# ===== MONITOR NEW =====
# UVM_INFO router_monitor.sv(25) @ 0: uvm_test_top.env.agt.mon [router_monitor] Inside router MONITOR build phase (L)
# =====
#
# ===== SEQUENCER BUILD=====
# UVM_INFO router_sequencer.sv(25) @ 0: uvm_test_top.env.agt.sqr [router_sequencer] Inside router SQR build phase (L)
# =====
#
# ===== COVERAGE NEW =====
# UVM_INFO router_coverage.sv(26) @ 0: uvm_test_top.env.cov [router_coverage] Inside router COVERAGE build phase (L)
# =====
#
# ===== SCOREBOARD BUILD =====
# UVM_INFO router_scoreboard.sv(24) @ 0: uvm_test_top.env.scb [router_scoreboard] Inside router SCOREBOARD build phase (L)
# =====
#
# --- UVM Report Summary ---

```

```

# --- UVM Report Summary ---
#
# ** Report counts by severity
# UVM_INFO : 18
# UVM_WARNING : 0
# UVM_ERROR : 0
# UVM_FATAL : 0
# ** Report counts by id
# [COVERAGE] 1
# [DRIVER] 1
# [ENV] 1
# [MONITOR] 1
# [RNTST] 1
# [SCOREBOARD] 1
# [SEQUENCE] 1
# [SQR] 1
# [TEST] 1
# [agent] 1
# [router_agent] 1
# [router_coverage] 1
# [router_driver] 1
# [router_env] 1
# [router_monitor] 1
# [router_scoreboard] 1
# [router_sequencer] 1
# [router_test] 1
# ** Note: $finish : D:/QuestaSim/win64/./verilog_src/uvm-1.1d/src/base/uvm_root.svh(430)
# Time: 0 ns Iteration: 215 Instance: /top
# 1
# Break in Task uvm_pkg/uvm_root::run_test at D:/QuestaSim/win64/./verilog_src/uvm-1.1d/src/base/uvm_root.svh line 430
VSIM 3>

```

2. Medium Verbosity

- DO file:

```
1 vlog -f router_files.txt
2 vsim -voptargs=+acc work.top +UVM_VERBOSITY=UVM_MEDIUM
3 run -all
```

- Log:

```
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(277) @ 0: reporter [Questa UVM] QUESTA_UVM-1.2.3
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(278) @ 0: reporter [Questa UVM] questa_uvm::init(+struct
# DEFAULT VERRBOSITY LEVEL IS 200
#
# ===== TEST NEW =====
# UVM_INFO router_test.sv(21) @ 0: uvm_test_top [TEST] creating new TEST class (L)
# UVM_INFO router_test.sv(22) @ 0: uvm_test_top [TEST] creating new TEST class (M)
# =====
#
# UVM_INFO @ 0: reporter [RNIST] Running test router_test...
#
# ===== ENVIROMENT NEW =====
# UVM_INFO router_enviroment.sv(25) @ 0: uvm_test_top.env [ENV] creating new ENV class (L)
# UVM_INFO router_enviroment.sv(26) @ 0: uvm_test_top.env [ENV] creating new ENV class (M)
# =====
#
# ===== SEQUENCE NEW =====
# UVM_INFO router_sequence.sv(14) @ 0: reporter@@router_sequence [SEQUENCE] creating new sequence class (L)
# UVM_INFO router_sequence.sv(15) @ 0: reporter@@router_sequence [SEQUENCE] creating new sequence class (M)
# =====
#
# ===== TEST BUILD =====
# UVM_INFO router_test.sv(35) @ 0: uvm_test_top [router_test] Inside router TEST build phase (L)
# UVM_INFO router_test.sv(36) @ 0: uvm_test_top [router_test] Inside router TEST build phase (M)
# =====
#
# ===== AGENT NEW =====
# UVM_INFO router_agent.sv(25) @ 0: uvm_test_top.env.agt [agent] creating new agent class (L)
# UVM_INFO router_agent.sv(26) @ 0: uvm_test_top.env.agt [agent] creating new agent class (M)
# =====
#
# ===== SCOREBOARD NEW =====
# UVM_INFO router_scoreboard.sv(13) @ 0: uvm_test_top.env.scb [SCOREBOARD] creating new SCOREBOARD class (L)
# UVM_INFO router_scoreboard.sv(14) @ 0: uvm_test_top.env.scb [SCOREBOARD] creating new SCOREBOARD class (M)
# =====
#
# ===== COVERAGE NEW =====
# UVM_INFO router_coverage.sv(14) @ 0: uvm_test_top.env.cov [COVERAGE] creating new COVERAGE class (L)
# UVM_INFO router_coverage.sv(15) @ 0: uvm_test_top.env.cov [COVERAGE] creating new COVERAGE class (M)
# =====
#
```

```

# ===== ENVIROMENT BUILD =====
# UVM_INFO router_enviroment.sv(39) @ 0: uvm_test_top.env [router_env] Inside router ENV build phase (L)
# UVM_INFO router_enviroment.sv(40) @ 0: uvm_test_top.env [router_env] Inside router ENV build phase (M)
# =====
#
#
# ===== SEQUENCER NEW =====
# UVM_INFO router_sequencer.sv(14) @ 0: uvm_test_top.env.agt.sqr [SQR] creating new SQR class (L)
# UVM_INFO router_sequencer.sv(15) @ 0: uvm_test_top.env.agt.sqr [SQR] creating new SQR class (M)
# =====
#
#
# ===== DRIVER NEW =====
# UVM_INFO router_driver.sv(13) @ 0: uvm_test_top.env.agt.drv [DRIVER] creating new DRIVER class (L)
# UVM_INFO router_driver.sv(14) @ 0: uvm_test_top.env.agt.drv [DRIVER] creating new DRIVER class (M)
# =====
#
#
# ===== MONITOR NEW =====
# UVM_INFO router_monitor.sv(13) @ 0: uvm_test_top.env.agt.mon [MONITOR] creating new MONITOR class (L)
# UVM_INFO router_monitor.sv(14) @ 0: uvm_test_top.env.agt.mon [MONITOR] creating new MONITOR class (M)
# =====
#
#
# ===== AGENT BUILD =====
# UVM_INFO router_agent.sv(40) @ 0: uvm_test_top.env.agt [router_agent] Inside router AGENT build phase (L)
# UVM_INFO router_agent.sv(41) @ 0: uvm_test_top.env.agt [router_agent] Inside router AGENT build phase (M)
# =====
#
#
# ===== DRIVER BUILD =====
# UVM_INFO router_driver.sv(25) @ 0: uvm_test_top.env.agt.drv [router_driver] Inside router DRIVER build phase (L)
# UVM_INFO router_driver.sv(26) @ 0: uvm_test_top.env.agt.drv [router_driver] Inside router DRIVER build phase (M)
# =====
#
#
# ===== MONITOR NEW =====
# UVM_INFO router_monitor.sv(25) @ 0: uvm_test_top.env.agt.mon [router_monitor] Inside router MONITOR build phase (L)
# UVM_INFO router_monitor.sv(26) @ 0: uvm_test_top.env.agt.mon [router_monitor] Inside router MONITOR build phase (M)
# =====
#
#
# ===== SEQUENCER BUILD=====
# UVM_INFO router_sequencer.sv(25) @ 0: uvm_test_top.env.agt.sqr [router_sequencer] Inside router SQR build phase (L)
# UVM_INFO router_sequencer.sv(26) @ 0: uvm_test_top.env.agt.sqr [router_sequencer] Inside router SQR build phase (M)
# =====

```

```

# ===== SEQUENCER BUILD=====
# UVM_INFO router_sequencer.sv(25) @ 0: uvm_test_top.env.agt.sqr [router_sequencer] Inside router SQR build phase (L)
# UVM_INFO router_sequencer.sv(26) @ 0: uvm_test_top.env.agt.sqr [router_sequencer] Inside router SQR build phase (M)
# =====
#
#
# ===== COVERAGE NEW =====
# UVM_INFO router_coverage.sv(26) @ 0: uvm_test_top.env.cov [router_coverage] Inside router COVERAGE build phase (L)
# UVM_INFO router_coverage.sv(27) @ 0: uvm_test_top.env.cov [router_coverage] Inside router COVERAGE build phase (M)
# =====
#
#
# ===== SCOREBOARD BUILD =====
# UVM_INFO router_scoreboard.sv(24) @ 0: uvm_test_top.env.scb [router_scoreboard] Inside router SCOREBOARD build phase (L)
# UVM_INFO router_scoreboard.sv(25) @ 0: uvm_test_top.env.scb [router_scoreboard] Inside router SCOREBOARD build phase (M)
# =====

```

```

#
#
# ===== COVERAGE NEW =====
# UVM_INFO router_coverage.sv(26) @ 0: uvm_test_top.env.cov [router_coverage] Inside router COVERAGE build phase (L)
# UVM_INFO router_coverage.sv(27) @ 0: uvm_test_top.env.cov [router_coverage] Inside router COVERAGE build phase (M)
# =====
#
#
# ===== SCOREBOARD BUILD =====
# UVM_INFO router_scoreboard.sv(24) @ 0: uvm_test_top.env.scb [router_scoreboard] Inside router SCOREBOARD build phase (L)
# UVM_INFO router_scoreboard.sv(25) @ 0: uvm_test_top.env.scb [router_scoreboard] Inside router SCOREBOARD build phase (M)
# =====
#
#
# --- UVM Report Summary ---
#
# ** Report counts by severity
# UVM_INFO : 37
# UVM_WARNING : 0
# UVM_ERROR : 0
# UVM_FATAL : 0
# ** Report counts by id
# [COVERAGE] 2
# [DRIVER] 2
# [ENV] 2
# [MONITOR] 2
# [Questa UVM] 2
# [RNTST] 1
# [SCOREBOARD] 2
# [SEQUENCE] 2
# [SQR] 2
# [TEST] 2
# [agent] 2
# [router_agent] 2
# [router_coverage] 2
# [router_driver] 2
# [router_env] 2
# [router_monitor] 2
# [router_scoreboard] 2
# [router_sequencer] 2
# [router_test] 2
# ** Note: $finish : D:/QuestaSim/win64/./verilog_src/uvm-1.1d/src/base/uvm_root.svh(430)
# Time: 0 ns Iteration: 215 Instance: /top
# 1
# Break in Task uvm_pkg/uvm_root::run_test at D:/QuestaSim/win64/./verilog_src/uvm-1.1d/src/base/uvm_root.svh line 430

```

3. High Verbosity

- DO file:

```

1 vlog -f router_files.txt
2 vsim -voptargs=+acc work.top +UVM_VERBOSITY=UVM_HIGH
3 run -all

```

- Log:

```
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(277) @ 0: reporter [Questa UVM] QUESTA_UVM-1.2.3
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(278) @ 0: reporter [Questa UVM] questa_uvm::init(+struct)
# DEFAULT VERRBOSITY LEVEL IS 300

# ===== TEST NEW =====
# UVM_INFO router_test.sv(21) @ 0: uvm_test_top [TEST] creating new TEST class (L)
# UVM_INFO router_test.sv(22) @ 0: uvm_test_top [TEST] creating new TEST class (M)
# UVM_INFO router_test.sv(23) @ 0: uvm_test_top [TEST] creating new TEST class (H)
# =====
#
# UVM_INFO @ 0: reporter [RNIST] Running test router_test...
#
# ===== ENVIROMENT NEW =====
# UVM_INFO router_enviroment.sv(25) @ 0: uvm_test_top.env [ENV] creating new ENV class (L)
# UVM_INFO router_enviroment.sv(26) @ 0: uvm_test_top.env [ENV] creating new ENV class (M)
# UVM_INFO router_enviroment.sv(27) @ 0: uvm_test_top.env [ENV] creating new ENV class (H)
# =====
#
# ===== SEQUENCE NEW =====
# UVM_INFO router_sequence.sv(14) @ 0: reporter@@router_sequence [SEQUENCE] creating new sequence class (L)
# UVM_INFO router_sequence.sv(15) @ 0: reporter@@router_sequence [SEQUENCE] creating new sequence class (M)
# UVM_INFO router_sequence.sv(16) @ 0: reporter@@router_sequence [SEQUENCE] creating new sequence class (H)
# =====
#
# ===== TEST BUILD =====
# UVM_INFO router_test.sv(35) @ 0: uvm_test_top [router_test] Inside router TEST build phase (L)
# UVM_INFO router_test.sv(36) @ 0: uvm_test_top [router_test] Inside router TEST build phase (M)
# UVM_INFO router_test.sv(37) @ 0: uvm_test_top [router_test] Inside router TEST build phase (H)
# =====
#
# ===== AGENT NEW =====
# UVM_INFO router_agent.sv(25) @ 0: uvm_test_top.env.agt [agent] creating new agent class (L)
# UVM_INFO router_agent.sv(26) @ 0: uvm_test_top.env.agt [agent] creating new agent class (M)
# UVM_INFO router_agent.sv(27) @ 0: uvm_test_top.env.agt [agent] creating new agent class (H)
# =====
#
# ===== SCOREBOARD NEW =====
# UVM_INFO router_scoreboard.sv(13) @ 0: uvm_test_top.env.scb [SCOREBOARD] creating new SCOREBOARD class (L)
# UVM_INFO router_scoreboard.sv(14) @ 0: uvm_test_top.env.scb [SCOREBOARD] creating new SCOREBOARD class (M)
# UVM_INFO router_scoreboard.sv(15) @ 0: uvm_test_top.env.scb [SCOREBOARD] creating new SCOREBOARD class (H)
# =====
#
# ===== COVERAGE NEW =====
# UVM_INFO router_coverage.sv(14) @ 0: uvm_test_top.env.cov [COVERAGE] creating new COVERAGE class (L)
# UVM_INFO router_coverage.sv(15) @ 0: uvm_test_top.env.cov [COVERAGE] creating new COVERAGE class (M)
# UVM_INFO router_coverage.sv(16) @ 0: uvm_test_top.env.cov [COVERAGE] creating new COVERAGE class (H)
# =====
#
# ===== ENVIROMENT BUILD =====
# UVM_INFO router_enviroment.sv(39) @ 0: uvm_test_top.env [router_env] Inside router ENV build phase (L)
# UVM_INFO router_enviroment.sv(40) @ 0: uvm_test_top.env [router_env] Inside router ENV build phase (M)
# UVM_INFO router_enviroment.sv(41) @ 0: uvm_test_top.env [router_env] Inside router ENV build phase (H)
# =====
#
# ===== SEQUENCER NEW =====
# UVM_INFO router_sequencer.sv(14) @ 0: uvm_test_top.env.agt.sqr [SQR] creating new SQR class (L)
# UVM_INFO router_sequencer.sv(15) @ 0: uvm_test_top.env.agt.sqr [SQR] creating new SQR class (M)
# UVM_INFO router_sequencer.sv(16) @ 0: uvm_test_top.env.agt.sqr [SQR] creating new SQR class (H)
# =====
```

```

# ===== DRIVER NEW =====
# UVM_INFO router_driver.sv(13) @ 0: uvm_test_top.env.agt.drv [DRIVER] creating new DRIVER class (L)
# UVM_INFO router_driver.sv(14) @ 0: uvm_test_top.env.agt.drv [DRIVER] creating new DRIVER class (M)
# UVM_INFO router_driver.sv(15) @ 0: uvm_test_top.env.agt.drv [DRIVER] creating new DRIVER class (H)
# =====
#
# ===== MONITOR NEW =====
# UVM_INFO router_monitor.sv(13) @ 0: uvm_test_top.env.agt.mon [MONITOR] creating new MONITOR class (L)
# UVM_INFO router_monitor.sv(14) @ 0: uvm_test_top.env.agt.mon [MONITOR] creating new MONITOR class (M)
# UVM_INFO router_monitor.sv(15) @ 0: uvm_test_top.env.agt.mon [MONITOR] creating new MONITOR class (H)
# =====
#
# ===== AGENT BUILD =====
# UVM_INFO router_agent.sv(40) @ 0: uvm_test_top.env.agt [router_agent] Inside router AGENT build phase (L)
# UVM_INFO router_agent.sv(41) @ 0: uvm_test_top.env.agt [router_agent] Inside router AGENT build phase (M)
# UVM_INFO router_agent.sv(42) @ 0: uvm_test_top.env.agt [router_agent] Inside router AGENT build phase (H)
# =====
#
# ===== DRIVER BUILD =====
# UVM_INFO router_driver.sv(25) @ 0: uvm_test_top.env.agt.drv [router_driver] Inside router DRIVER build phase (L)
# UVM_INFO router_driver.sv(26) @ 0: uvm_test_top.env.agt.drv [router_driver] Inside router DRIVER build phase (M)
# UVM_INFO router_driver.sv(27) @ 0: uvm_test_top.env.agt.drv [router_driver] Inside router DRIVER build phase (H)
# =====
#
# ===== MONITOR NEW =====
# UVM_INFO router_monitor.sv(25) @ 0: uvm_test_top.env.agt.mon [router_monitor] Inside router MONITOR build phase (L)
# UVM_INFO router_monitor.sv(26) @ 0: uvm_test_top.env.agt.mon [router_monitor] Inside router MONITOR build phase (M)
# UVM_INFO router_monitor.sv(27) @ 0: uvm_test_top.env.agt.mon [router_monitor] Inside router MONITOR build phase (H)
# =====
#
# ===== SEQUENCER BUILD=====
# UVM_INFO router_sequencer.sv(25) @ 0: uvm_test_top.env.agt.sqr [router_sequencer] Inside router SQR build phase (L)
# UVM_INFO router_sequencer.sv(26) @ 0: uvm_test_top.env.agt.sqr [router_sequencer] Inside router SQR build phase (M)
# UVM_INFO router_sequencer.sv(27) @ 0: uvm_test_top.env.agt.sqr [router_sequencer] Inside router SQR build phase (H)
# =====
#
# ===== COVERAGE NEW =====
# UVM_INFO router_coverage.sv(26) @ 0: uvm_test_top.env.cov [router_coverage] Inside router COVERAGE build phase (L)
# UVM_INFO router_coverage.sv(27) @ 0: uvm_test_top.env.cov [router_coverage] Inside router COVERAGE build phase (M)
# UVM_INFO router_coverage.sv(28) @ 0: uvm_test_top.env.cov [router_coverage] Inside router COVERAGE build phase (H)
# =====
#
# ===== SCOREBOARD BUILD =====
# UVM_INFO router_scoreboard.sv(24) @ 0: uvm_test_top.env.scb [router_scoreboard] Inside router SCOREBOARD build phase (L)
# UVM_INFO router_scoreboard.sv(25) @ 0: uvm_test_top.env.scb [router_scoreboard] Inside router SCOREBOARD build phase (M)
# UVM_INFO router_scoreboard.sv(26) @ 0: uvm_test_top.env.scb [router_scoreboard] Inside router SCOREBOARD build phase (H)
# =====
#
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(339) @ 0: reporter [Questa UVM] End Of Elaboration
#
# --- UVM Report Summary ---

```

```

# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(339) @ 0: reporter [Questa UVM] End Of Elaboration
#
# --- UVM Report Summary ---
#
# ** Report counts by severity
# UVM_INFO :    55
# UVM_WARNING :    0
# UVM_ERROR :    0
# UVM_FATAL :    0
# ** Report counts by id
# [COVERAGE]    3
# [DRIVER]      3
# [ENV]         3
# [MONITOR]     3
# [Questa UVM]  3
# [RNTST]       1
# [SCOREBOARD]  3
# [SEQUENCE]    3
# [SQR]         3
# [TEST]        3
# [agent]       3
# [router_agent] 3
# [router_coverage] 3
# [router_driver] 3
# [router_env]   3
# [router_monitor] 3
# [router_scoreboard] 3
# [router_sequencer] 3
# [router_test]  3
# ** Note: $finish : D:/QuestaSim/win64/./verilog_src/uvm-1.1d/src/base/uvm_root.svh(430)
# Time: 0 ns Iteration: 215 Instance: /top
# 1
# Break in Task uvm_pkg/uvm_root::run_test at D:/QuestaSim/win64/./verilog_src/uvm-1.1d/src/base/uvm_root.svh line 430

```

4. Full Verbosity

- DO file:

```

1 vlog -f router_files.txt
2 vsim -voptargs=+acc work.top +UVM_VERBOSITY=UVM_FULL
3 run -all

```

- Log:

```

-
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(277) @ 0: reporter [Questa UVM] QUESTA_UVM-1.2.3
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(278) @ 0: reporter [Questa UVM] questa_uvm::init(+struct)
# DEFAULT VERRBOSITY LEVEL IS 400
#
# ===== TEST NEW =====
# UVM_INFO router_test.sv(21) @ 0: uvm_test_top [TEST] creating new TEST class (L)
# UVM_INFO router_test.sv(22) @ 0: uvm_test_top [TEST] creating new TEST class (M)
# UVM_INFO router_test.sv(23) @ 0: uvm_test_top [TEST] creating new TEST class (H)
# UVM_INFO router_test.sv(24) @ 0: uvm_test_top [TEST] creating new TEST class (FULL)
# =====

```



```

# ===== SEQUENCER NEW =====
# UVM_INFO router_sequencer.sv(14) @ 0: uvm_test_top.env.agt.sqr [SQR] creating new SQR class (L)
# UVM_INFO router_sequencer.sv(15) @ 0: uvm_test_top.env.agt.sqr [SQR] creating new SQR class (M)
# UVM_INFO router_sequencer.sv(16) @ 0: uvm_test_top.env.agt.sqr [SQR] creating new SQR class (H)
# UVM_INFO router_sequencer.sv(17) @ 0: uvm_test_top.env.agt.sqr [SQR] creating new SQR class (FULL)
# =====
#
# ===== DRIVER NEW =====
# UVM_INFO router_driver.sv(13) @ 0: uvm_test_top.env.agt.drv [DRIVER] creating new DRIVER class (L)
# UVM_INFO router_driver.sv(14) @ 0: uvm_test_top.env.agt.drv [DRIVER] creating new DRIVER class (M)
# UVM_INFO router_driver.sv(15) @ 0: uvm_test_top.env.agt.drv [DRIVER] creating new DRIVER class (H)
# UVM_INFO router_driver.sv(16) @ 0: uvm_test_top.env.agt.drv [DRIVER] creating new DRIVER class (FULL)
# =====
#
# ===== MONITOR NEW =====
# UVM_INFO router_monitor.sv(13) @ 0: uvm_test_top.env.agt.mon [MONITOR] creating new MONITOR class (L)
# UVM_INFO router_monitor.sv(14) @ 0: uvm_test_top.env.agt.mon [MONITOR] creating new MONITOR class (M)
# UVM_INFO router_monitor.sv(15) @ 0: uvm_test_top.env.agt.mon [MONITOR] creating new MONITOR class (H)
# UVM_INFO router_monitor.sv(16) @ 0: uvm_test_top.env.agt.mon [MONITOR] creating new MONITOR class (FULL)
# =====
#
# ===== AGENT BUILD =====
# UVM_INFO router_agent.sv(40) @ 0: uvm_test_top.env.agt [router_agent] Inside router AGENT build phase (L)
# UVM_INFO router_agent.sv(41) @ 0: uvm_test_top.env.agt [router_agent] Inside router AGENT build phase (M)
# UVM_INFO router_agent.sv(42) @ 0: uvm_test_top.env.agt [router_agent] Inside router AGENT build phase (H)
# UVM_INFO router_agent.sv(43) @ 0: uvm_test_top.env.agt [router_agent] Inside router AGENT build phase (FULL)
# =====
#
# ===== DRIVER BUILD =====
# UVM_INFO router_driver.sv(25) @ 0: uvm_test_top.env.agt.drv [router_driver] Inside router DRIVER build phase (L)
# UVM_INFO router_driver.sv(26) @ 0: uvm_test_top.env.agt.drv [router_driver] Inside router DRIVER build phase (M)
# UVM_INFO router_driver.sv(27) @ 0: uvm_test_top.env.agt.drv [router_driver] Inside router DRIVER build phase (H)
# UVM_INFO router_driver.sv(28) @ 0: uvm_test_top.env.agt.drv [router_driver] Inside router DRIVER build phase (FULL)
# =====
#
# ===== MONITOR BUILD =====
# UVM_INFO router_monitor.sv(25) @ 0: uvm_test_top.env.agt.mon [router_monitor] Inside router MONITOR build phase (L)
# UVM_INFO router_monitor.sv(26) @ 0: uvm_test_top.env.agt.mon [router_monitor] Inside router MONITOR build phase (M)
# UVM_INFO router_monitor.sv(27) @ 0: uvm_test_top.env.agt.mon [router_monitor] Inside router MONITOR build phase (H)
# UVM_INFO router_monitor.sv(28) @ 0: uvm_test_top.env.agt.mon [router_monitor] Inside router MONITOR build phase (FULL)
# =====
#
# ===== SEQUENCER BUILD =====
# UVM_INFO router_sequencer.sv(25) @ 0: uvm_test_top.env.agt.sqr [router_sequencer] Inside router SQR build phase (L)
# UVM_INFO router_sequencer.sv(26) @ 0: uvm_test_top.env.agt.sqr [router_sequencer] Inside router SQR build phase (M)
# UVM_INFO router_sequencer.sv(27) @ 0: uvm_test_top.env.agt.sqr [router_sequencer] Inside router SQR build phase (H)
# UVM_INFO router_sequencer.sv(28) @ 0: uvm_test_top.env.agt.sqr [router_sequencer] Inside router SQR build phase (FULL)
# =====
#
# ===== COVERAGE NEW =====
# UVM_INFO router_coverage.sv(26) @ 0: uvm_test_top.env.cov [router_coverage] Inside router COVERAGE build phase (L)
# UVM_INFO router_coverage.sv(27) @ 0: uvm_test_top.env.cov [router_coverage] Inside router COVERAGE build phase (M)
# UVM_INFO router_coverage.sv(28) @ 0: uvm_test_top.env.cov [router_coverage] Inside router COVERAGE build phase (H)
# UVM_INFO router_coverage.sv(29) @ 0: uvm_test_top.env.cov [router_coverage] Inside router COVERAGE build phase (FULL)
# =====
#
# ===== SCOREBOARD BUILD =====
# UVM_INFO router_scoreboard.sv(24) @ 0: uvm_test_top.env.scb [router_scoreboard] Inside router SCOREBOARD build phase (L)
# UVM_INFO router_scoreboard.sv(25) @ 0: uvm_test_top.env.scb [router_scoreboard] Inside router SCOREBOARD build phase (M)
# UVM_INFO router_scoreboard.sv(26) @ 0: uvm_test_top.env.scb [router_scoreboard] Inside router SCOREBOARD build phase (H)
# UVM_INFO router_scoreboard.sv(27) @ 0: uvm_test_top.env.scb [router_scoreboard] Inside router SCOREBOARD build phase (FULL)
# =====

```

♀ Note: the following Info wasn't printed with the previous verbosity levels (Lower levels than FULL)

```
# UVM_INFO verilog_src/questa_uvm_pkg-1.2/src/questa_uvm_pkg.sv(339) @ 0: reporter [Questa UVM] End Of Elaboration
# UVM_INFO verilog_src/uvm-1.1d/src/seq/uvm_sequencer_base.svh(1384) @ 0: uvm_test_top.env.agt.sqr [PHASESEQ] No default phase sequence for phase 'run'
# UVM_INFO verilog_src/uvm-1.1d/src/seq/uvm_sequencer_base.svh(1384) @ 0: uvm_test_top.env.agt.sqr [PHASESEQ] No default phase sequence for phase 'pre_reset'
# UVM_INFO verilog_src/uvm-1.1d/src/seq/uvm_sequencer_base.svh(1384) @ 0: uvm_test_top.env.agt.sqr [PHASESEQ] No default phase sequence for phase 'reset'
# UVM_INFO verilog_src/uvm-1.1d/src/seq/uvm_sequencer_base.svh(1384) @ 0: uvm_test_top.env.agt.sqr [PHASESEQ] No default phase sequence for phase 'post_reset'
# UVM_INFO verilog_src/uvm-1.1d/src/seq/uvm_sequencer_base.svh(1384) @ 0: uvm_test_top.env.agt.sqr [PHASESEQ] No default phase sequence for phase 'pre_configure'
# UVM_INFO verilog_src/uvm-1.1d/src/seq/uvm_sequencer_base.svh(1384) @ 0: uvm_test_top.env.agt.sqr [PHASESEQ] No default phase sequence for phase 'configure'
# UVM_INFO verilog_src/uvm-1.1d/src/seq/uvm_sequencer_base.svh(1384) @ 0: uvm_test_top.env.agt.sqr [PHASESEQ] No default phase sequence for phase 'post_configure'
# UVM_INFO verilog_src/uvm-1.1d/src/seq/uvm_sequencer_base.svh(1384) @ 0: uvm_test_top.env.agt.sqr [PHASESEQ] No default phase sequence for phase 'main'
# UVM_INFO verilog_src/uvm-1.1d/src/seq/uvm_sequencer_base.svh(1384) @ 0: uvm_test_top.env.agt.sqr [PHASESEQ] No default phase sequence for phase 'post_main'
# UVM_INFO verilog_src/uvm-1.1d/src/seq/uvm_sequencer_base.svh(1384) @ 0: uvm_test_top.env.agt.sqr [PHASESEQ] No default phase sequence for phase 'pre_shutdown'
# UVM_INFO verilog_src/uvm-1.1d/src/seq/uvm_sequencer_base.svh(1384) @ 0: uvm_test_top.env.agt.sqr [PHASESEQ] No default phase sequence for phase 'shutdown'
# UVM_INFO verilog_src/uvm-1.1d/src/seq/uvm_sequencer_base.svh(1384) @ 0: uvm_test_top.env.agt.sqr [PHASESEQ] No default phase sequence for phase 'post_shutdown'
#
# --- UVM Report Summary ---
```

Then, normal UVM summary

```
# --- UVM Report Summary ---
#
# ** Report counts by severity
# UVM_INFO : 85
# UVM_WARNING : 0
# UVM_ERROR : 0
# UVM_FATAL : 0
# ** Report counts by id
# [COVERAGE] 4
# [DRIVER] 4
# [ENV] 4
# [MONITOR] 4
# [PHASESEQ] 13
# [Questa UVM] 3
# [RNTST] 1
# [SCOREBOARD] 4
# [SEQUENCE] 4
# [SQR] 4
# [TEST] 4
# [agent] 4
# [router_agent] 4
# [router_coverage] 4
# [router_driver] 4
# [router_env] 4
# [router_monitor] 4
# [router_scoreboard] 4
# [router_sequencer] 4
# [router_test] 4
# ** Note: $finish : D:/QuestaSim/win64/./verilog_src/uvm-1.1d/src/base/uvm_root.svh(430)
# Time: 0 ns Iteration: 215 Instance: /top
# 1
# Break in Task uvm_pkg/uvm_root::run_test at D:/QuestaSim/win64/./verilog_src/uvm-1.1d/src/base/uvm_root.svh line 430
VSIM 3>
```